

The key idea underlying this simulation technique is the systematic translation of formulas into basic sentences. Suppose  $\vdash \varphi$  and  $\varphi$  contains a subformula  $a \wedge b$ , where  $a, b \in \mathbb{B}$ . Following the natural deduction rules governing conjunction, Sandqvist makes sure to include the following rules in the “specifically tailored” base  $\mathcal{N}$  for  $\varphi$ :

$$\frac{a \quad b}{r} \quad \frac{r}{a} \quad \frac{r}{b}$$

where  $r$  is a fresh basic sentence representing  $a \wedge b$ . This means that  $r$  behaves in  $\mathcal{N}$  as  $a \wedge b$  behaves in Gentzen’s NJ [1] — that is, as in intuitionistic sentential logic.

More generally, each subformula  $\chi$  of  $\varphi$  is assigned a corresponding basic counterpart  $\chi^b$ , e.g.,  $r = (a \wedge b)^b$ . Sandqvist establishes their equivalence within  $\mathcal{N}$ :

$$\chi \Vdash_{\mathcal{N}} \chi^b \quad \text{and} \quad \chi^b \Vdash_{\mathcal{N}} \chi.$$

Since we assume  $\Vdash \varphi$ , it follows that  $\Vdash \varphi^b$ , and given that every rule in  $\mathcal{N}$  corresponds to an intuitionistic natural deduction rule, we conclude  $\vdash \varphi$ , as required.

The *flattening* technique employed here is strikingly different from standard completeness proofs. However, as the saying goes, “*all theorems were already proved in the Soviet Union*”. In particular, this approach bears a strong resemblance to a method developed by Mints [11, 12, 13, 14, 15] from the 1980s for evaluating intuitionistic consequence via resolution systems. Mints’ approach systematically transforms proof systems for various logics, including intuitionistic logic, into resolution-based systems while preserving derivational structure. This extends earlier work by Maslov [10] on resolution calculi for classical predicate logic.

Mints defines a class of formulas, *clauses*, with a particularly simple structure (details below). Given a formula  $\varphi$ , he constructs a corresponding set of clauses  $M$  such that for some designated basic sentence  $g$ ,

$$\vdash \varphi \quad \text{iff} \quad M \vdash g \tag{\dagger}$$

Each subformula  $\chi$  of  $\varphi$  is associated with an basic counterpart  $\chi^b$ , and  $M$  is constructed by adding clauses ensuring that  $\chi^b$  correctly encodes the logical behaviour of  $\chi$ . For instance, if  $\chi = \chi_1 \wedge \chi_2$ , the following clauses are introduced:

$$\chi_1^b \wedge \chi_2^b \rightarrow \chi^b, \quad \chi^b \rightarrow \chi_1^b, \quad \chi^b \rightarrow \chi_2^b.$$

The key observation is that relative to such clauses,

$$(\chi_1 \wedge \chi_2)^b \leftrightarrow \chi_1^b \wedge \chi_2^b.$$

Thus, reasoning about  $\varphi$  reduces to reasoning about formulas with at most three atomic components. In  $(\dagger)$ ,  $M$  is the clause set encoding  $\varphi$  and  $g = \varphi^b$ .

A natural correspondence emerges between bases and clausal systems. Specifically, for any formula  $\varphi$ , Sandqvist’s  $\mathcal{N}$  and Mints’  $M$  coincide when identical atomic names are used for subformulas. More significantly, this bijection establishes a pointwise correspondence between support in a base and proof search in a clause set, aligning validity in B-eS for  $\varphi$  with resolution in  $M$  for  $g$ . Consequently, the soundness and completeness of intuitionistic logic with respect to Sandqvist’s B-eS *both* follow as corollaries of Mints’ theorem.